WHEN SELF-SUPERVISION FAILS: UNMASKING DIFFERENTIAL SHORTCUT LEARNING IN CONTRASTIVE AND MASKED AUTOENCODING METHODS

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ABSTRACT

Self-supervised learning (SSL) has become the dominant paradigm for pre-training foundation models, yet how different SSL objectives learn spurious correlations remains poorly understood. This work provides the first systematic comparison of spurious correlation learning across contrastive learning (SimCLR) and masked image modeling (MAE) using controlled datasets with known spurious correlations. We evaluate representation quality via linear probing and robustness via worst-group accuracy on datasets with shifted correlations. Our key finding reveals a surprising reversal of expectations: contrary to hypotheses that contrastive methods are more vulnerable to global texture shortcuts, we find that MAE exhibits substantially worse robustness to both background patterns and local spurious features, achieving worst-group accuracies of 5.2% and 38.8% compared to SimCLR's 24.0% and 68.0% on CIFAR10-BG and Fashion-Pattern respectively. Both methods completely fail on color-based spurious correlations (0% worst-group accuracy). Through extensive ablations, we demonstrate that data augmentation is the most critical component for robustness, while architectural choices show dataset-dependent effects. This work provides foundational understanding of SSL inductive biases and highlights the urgent need for targeted interventions to address color-based shortcuts in self-supervised pre-training.

1 INTRODUCTION

Self-supervised learning has revolutionized representation learning, with methods like SimCLR (Chen et al., 2020), MAE (He et al., 2021), and MoCo (He et al., 2019) achieving remarkable performance across diverse downstream tasks. However, a critical question remains: how do different SSL objectives learn spurious correlations during pre-training? This question is important because spurious correlations learned during pre-training propagate to all downstream tasks (Mehta et al., 2022), yet no systematic comparison exists across SSL paradigms.

Prior work has extensively studied spurious correlations in supervised learning (Sagawa et al., 2020a;b), demonstrating that overparameterized models can achieve high average accuracy while catastrophically failing on minority groups. Recent work shows that pre-trained representations can mitigate some spurious correlations (Mehta et al., 2022), but the mechanisms by which different SSL objectives encode shortcuts remain unclear. Moreover, CNNs exhibit strong texture bias (Geirhos et al., 2018), suggesting texture-based spurious correlations may be particularly problematic.

We hypothesized that contrastive methods would be more vulnerable to global texture shortcuts due to their instance discrimination objective, while masked modeling methods would be more susceptible to local pattern shortcuts due to their pixel-level reconstruction focus. To test this, we construct three controlled datasets with different types of spurious correlations: color-label correlations (Color-MNIST), background pattern correlations (CIFAR10-BG), and local pattern correlations (Fashion-Pattern). We pre-train SimCLR and MAE models and evaluate them using worst-group accuracy (Sagawa et al., 2020a) on test sets with shifted correlations. Our findings reveal that MAE consistently shows worse robustness than SimCLR across background and local pattern spurious correlations, achieving 2–5× lower worst-group accuracy. Both methods completely fail on color-based spurious correlations. Through extensive ablations on augmentation, architectural components, and hyperparameters, we identify data augmentation as the most critical factor for robustness.

2 Related Work

Spurious Correlations in Supervised Learning. Sagawa et al. (Sagawa et al., 2020a) introduced group DRO to handle spurious correlations in supervised learning, establishing worst-group accuracy as the standard metric for robustness. Follow-up work (Sagawa et al., 2020b) showed that overparameterization exacerbates spurious correlation learning. MaskTune (Taghanaki et al., 2022) proposed masking strategies to mitigate shortcuts without supervision. However, these works focus on supervised learning, while our work investigates how SSL objectives inherently differ in shortcut learning.

Self-Supervised Learning Methods. Contrastive learning methods like SimCLR (Chen et al., 2020) and MoCo (He et al., 2019) learn by maximizing agreement between augmented views. Masked image modeling approaches like MAE (He et al., 2021) and SimMIM (Xie et al., 2021) learn by reconstructing masked image patches. JEPA (Assran et al., 2023) uses predictive coding in representation space. While these methods achieve strong downstream performance, their differential susceptibilities to spurious correlations remain unexplored.

Robustness of Pre-trained Representations. Recent work shows that pre-trained representations can improve worst-group accuracy (Mehta et al., 2022), and that SSL features exhibit different robustness properties than supervised features (Mehlman & Narayanan, 2025). However, systematic comparisons across SSL paradigms are lacking.

3 Method

SSL Training Objectives. We implement SimCLR (Chen et al., 2020) with InfoNCE contrastive loss: given two augmented views x_i^1, x_i^2 of image x_i, we maximize agreement between their projections while contrasting with other images. For MAE (He et al., 2021), we mask random patches and train the model to reconstruct the original image using MSE loss. Both methods use convolutional encoders adapted to our dataset resolutions.

Evaluation Protocol. Following standard practice (Grill et al., 2020), we freeze pre-trained encoders and train linear classifiers on the learned representations. We measure: (1) average test accuracy across all groups, (2) worst-group accuracy on minority groups where spurious correlations are broken, and (3) Spurious Feature Reliance Score (SFRS) defined as $(acc_{avg} - acc_{worst})/acc_{avg}$, where lower values indicate better robustness.

Datasets with Controlled Spurious Correlations. We construct three datasets: *Color-MNIST* applies color spuriously correlated with digit class (digits 0–4 vs 5–9); *CIFAR10-BG* adds background patterns (checkerboard vs dots) correlated with class; *Fashion-Pattern* places local pattern stamps (top-left vs bottom-right) correlated with fashion item category. Training sets have 95% spurious correlation, while test sets have 50% majority group (correlation holds) and 50% minority group (correlation broken).

4 Experimental Setup

We use convolutional encoders with 128-dimensional embeddings. SimCLR uses a 2-layer projection head with 64-dimensional output and temperature $\tau = 0.5$. MAE uses a convolutional decoder. We pre-train for 15 epochs with Adam optimizer (lr=1e-3, batch size=64). Data augmentations include random horizontal flips and Gaussian noise ($\sigma = 0.05$). Linear probes are trained for 50 epochs with L-BFGS. We conduct systematic ablations on: data augmentation presence, projection head architecture, feature normalization, embedding dimensionality (32, 64, 128, 256), temperature scaling (0.5 vs 1.0), MAE decoder complexity, MAE masking ratio (0%, 50%, 75%), pooling strategy, and augmentation stochasticity.

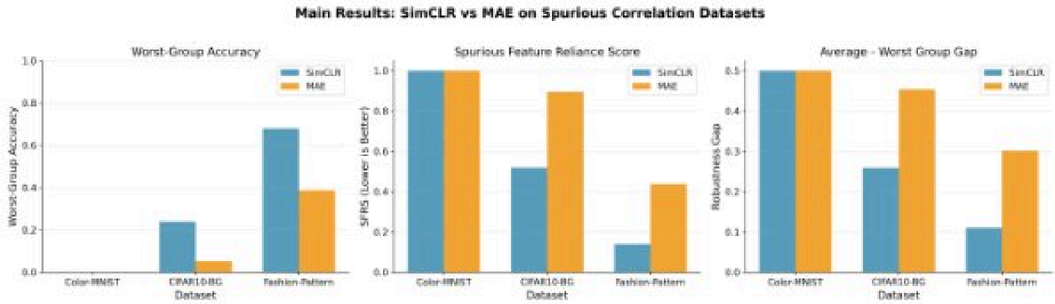


Figure 1: Main results comparing SimCLR and MAE across three spurious correlation datasets. Left: Worst-group accuracy (higher is better). Middle: Spurious Feature Reliance Score (lower is better). Right: Robustness gap between average and worst-group accuracy (lower is better). SimCLR demonstrates superior robustness on CIFAR10-BG and Fashion-Pattern, while both methods fail completely on Color-MNIST.

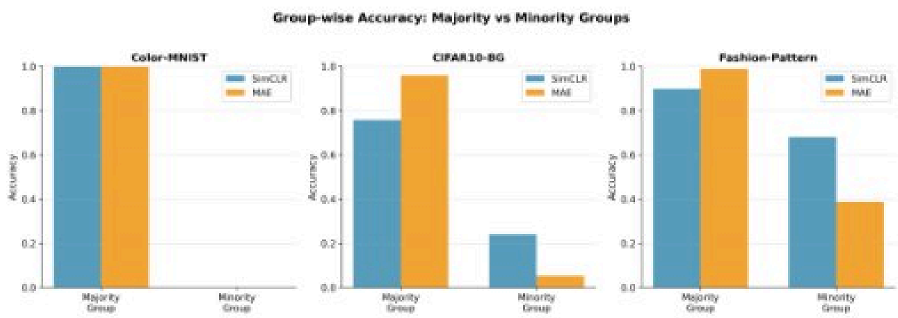


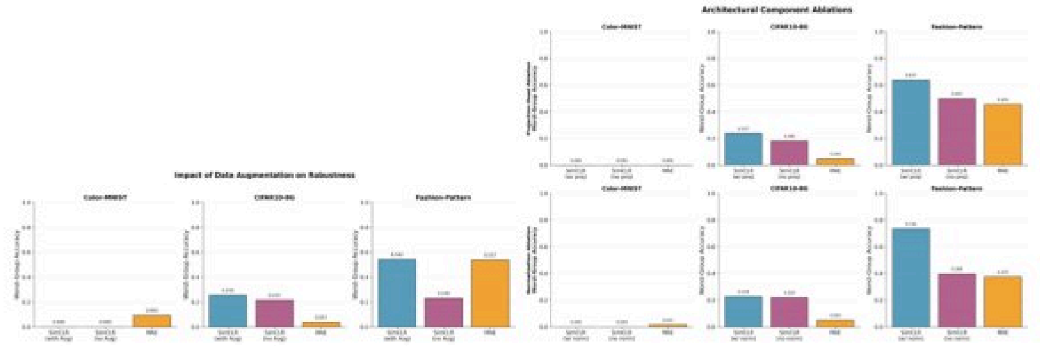
Figure 2: Group-wise accuracy breakdown showing majority vs minority group performance. Both methods completely fail on Color-MNIST minority groups. For CIFAR10-BG and Fashion-Pattern, MAE achieves higher majority accuracy but suffers catastrophic minority group failure, while SimCLR demonstrates more balanced performance across groups.

5 Results

Main Results. Figure 1 presents our key findings. On Color-MNIST, both methods achieve perfect majority group accuracy (100%) but completely fail on the minority group (0%), indicating total reliance on the color spurion. On CIFAR10-BG, SimCLR achieves 24.0% worst-group accuracy compared to MAE's 5.2%, demonstrating substantially better robustness to background patterns. On Fashion-Pattern, SimCLR achieves 68.0% worst-group accuracy versus MAE's 38.8%. These results contradict our initial hypothesis that contrastive methods would be more vulnerable to global shortcuts.

Group-wise Performance. Figure 2 shows the stark disparity between majority and minority groups across all three datasets. On Color-MNIST, both methods achieve near-perfect majority group accuracy but 0% minority accuracy, representing a qualitatively different failure mode. MAE achieves near-perfect majority group accuracy (96–100%) on CIFAR10-BG and Fashion-Pattern but exhibits severe minority group collapse (5.2% and 38.8%). SimCLR shows more balanced performance across groups. The SFRS metric quantifies this: SimCLR achieves 0.519 and 0.139 on CIFAR10-BG and Fashion-Pattern respectively, compared to MAE's 0.896 and 0.437.

Impact of Data Augmentation. Our ablation studies (Figure 3a) reveal that data augmentation is the most critical component for robustness. On Fashion-Pattern, removing augmentation from SimCLR reduces worst-group accuracy from 54.2% to 23.0%, a 31.2% absolute drop. On CIFAR10-BG, augmentation provides a 4.0% improvement for SimCLR. Interestingly, MAE shows minimal sensitivity to augmentation, suggesting its reconstruction objective may inherently encode spurious features regardless of augmentation strategy.



(a) Data Augmentation (SimCLR) (b) Architectural Components

Figure 3: Key ablation results. (a) Data augmentation is critical for SimCLR robustness, providing 31.2% improvement on Fashion-Pattern. (b) Projection head and normalization show dataset-dependent effects, with largest benefits on Fashion-Pattern.

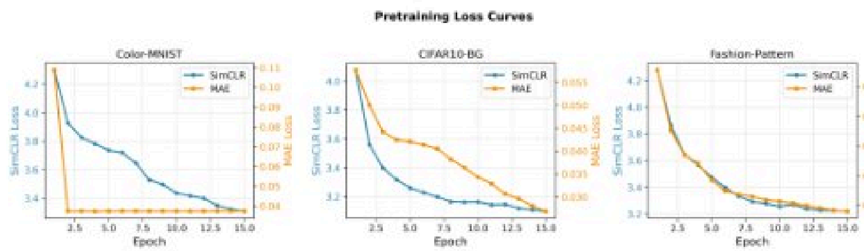


Figure 4: Pretraining loss curves showing SimCLR's gradual convergence versus MAE's rapid plateau. MAE quickly reaches near-zero losses within the first few epochs, suggesting the reconstruction task may be too simple, potentially contributing to spurious feature encoding.

Architectural and Hyperparameter Effects. The projection head provides modest improvements for SimCLR (1.5–14% across datasets), while normalization shows strong dataset-dependent effects (33.7% improvement on Fashion-Pattern, minimal on others). Embedding dimensionality shows non-monotonic effects: 256-dim is best for CIFAR10-BG while 64-dim is optimal for Fashion-Pattern. Temperature scaling provides slight improvements on CIFAR10-BG (3%) but degrades Fashion-Pattern performance (11%). For MAE, increasing masking ratio to 75% provides modest improvements (1.3–8.8%), while decoder complexity shows minimal impact.

Pretraining Dynamics. Figure 4 shows that SimCLR losses decrease gradually from $\sim$4.1–4.3 to $\sim$3.1–3.3 over 15 epochs, indicating stable convergence. MAE losses start at $\sim$0.09–0.12 and quickly reach near-zero ($\sim$0.03–0.05) within the first few epochs, then plateau. This rapid convergence may contribute to MAE's tendency to encode spurious low-level cues.

6 Discussion and Limitations

Our results reveal that MAE is substantially more vulnerable to spurious correlations than SimCLR on background patterns and local features, contrary to our initial hypothesis. We attribute this to MAE's pixel-level reconstruction objective, which may overfit to easy-to-predict local cues present in the majority group. SimCLR's augmentation-based invariance and instance discrimination appear to provide better robustness, though neither method handles color-based spurious correlations. The complete failure of both methods on Color-MNIST (0% worst-group accuracy) represents a critical unsolved challenge. Standard augmentations (flips, noise) preserve color information, allowing models to exploit this shortcut. This suggests that color-based spurious correlations require targeted interventions beyond standard SSL practices.

Our study has several limitations. First, we use synthetic spurious correlations that may not capture real-world complexity, though we validate findings across multiple spurious types. Second, we focus on small-scale experiments due to computational constraints, using shorter training (15 vs 800+ epochs) and smaller architectures than typical SSL deployments. Third, we evaluate only two SSL paradigms (contrastive and masked modeling), leaving predictive coding methods like JEPA (Assran et al., 2023) for future work.

7 Conclusion

This work provides the first systematic comparison of spurious correlation learning across SSL paradigms. Our key findings are: (1) MAE exhibits substantially worse robustness than SimCLR on background and local pattern spurious correlations, with 2–5$\times$ lower worst-group accuracy; (2) both methods completely fail on color-based spurious correlations, highlighting a critical unsolved challenge; (3) data augmentation is the most impactful component for robustness, while architectural choices show dataset-dependent effects. These results challenge common assumptions about SSL robustness and highlight the need for targeted interventions, particularly for color-based shortcuts. Future work should investigate predictive coding methods, scale to larger datasets and longer training, and develop SSL-specific augmentation strategies that explicitly target spurious correlation types identified in this study.

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SUPPLEMENTARY MATERIAL

A ADDITIONAL ABLATION RESULTS

Figure 5 shows embedding dimensionality and temperature scaling effects. Embedding dimensionality exhibits non-monotonic behavior: performance generally improves from 32 to 128 dimensions, but 256 dimensions show mixed results. Temperature scaling (T=1.0 vs baseline T=0.5) provides modest improvements on CIFAR10-BG but degrades Fashion-Pattern performance.

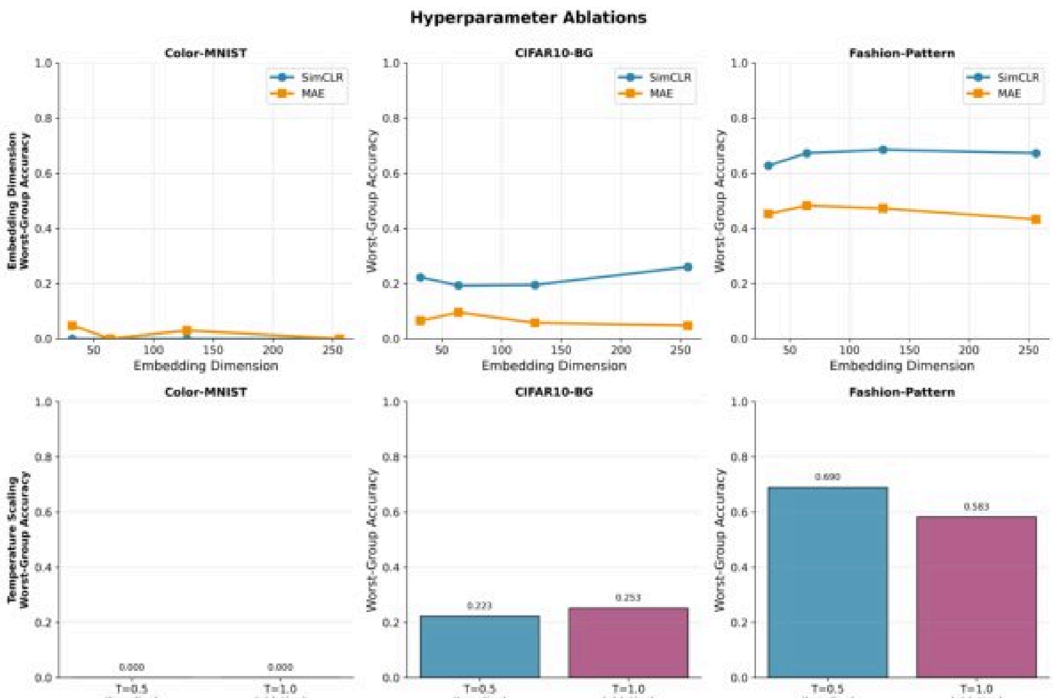

Figure 5: Hyperparameter ablations showing SimCLR performance. Top: Embedding dimensionality shows non-monotonic effects with dataset-dependent optima. Bottom: Temperature scaling (T=1.0 vs T=0.5) provides slight improvements on CIFAR10-BG (from 23.3% to 25.3%) but harms Fashion-Pattern performance (from 69.0% to 58.3%).

Figure 6 presents MAE-specific ablations. Decoder complexity shows mixed effects: minimal changes on Color-MNIST and CIFAR10-BG, but substantial degradation on Fashion-Pattern (11.3% drop from 44.8% to 33.5%). Masking ratio ablations reveal that higher masking (75%) provides modest improvements over no masking (1.3–8.8% across datasets), though 50% masking shows inconsistent results, sometimes underperforming the baseline.

B CROSS-DATASET SUMMARY

Figure 7 provides a comprehensive cross-dataset performance summary using heatmaps. The worst-group accuracy heatmap clearly shows SimCLR's superior robustness on CIFAR10-BG and Fashion-Pattern, while both methods fail identically on Color-MNIST. The SFRS heatmap confirms that SimCLR exhibits lower spurious feature reliance across all non-trivial datasets. The robustness

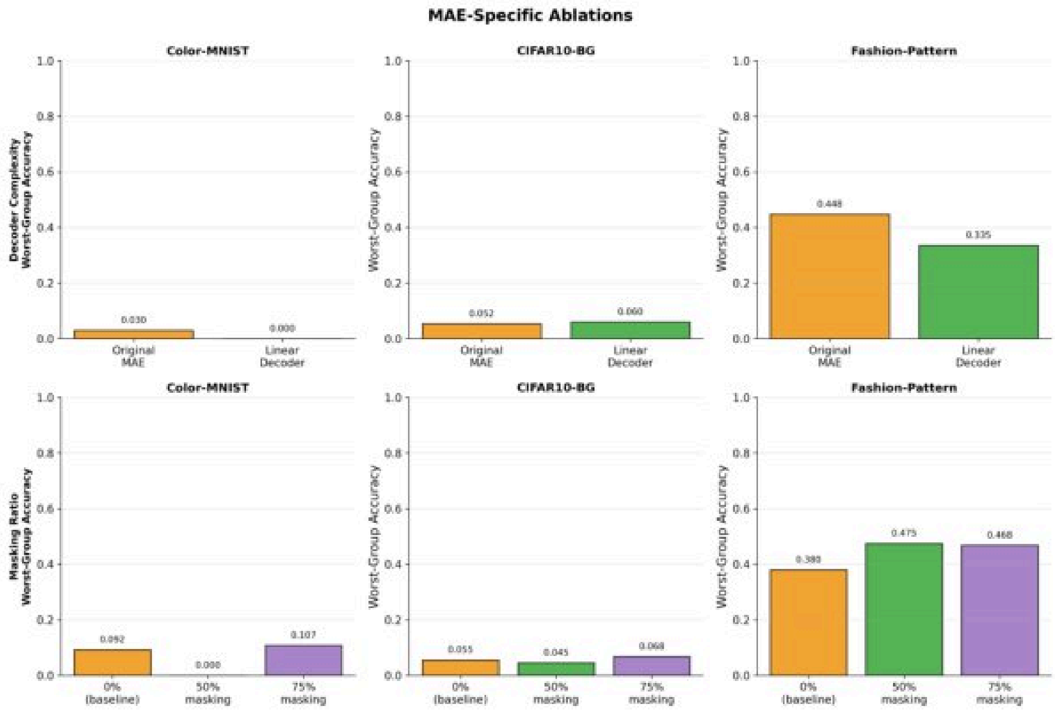

Figure 6: MAE-specific ablations. Top: Decoder complexity shows mixed effects, with substantial degradation on Fashion-Pattern (11.3% drop) but minimal changes on other datasets. Bottom: Higher masking ratios (75%) provide modest robustness improvements (1.3–8.8% across datasets), though 50% masking shows non-monotonic behavior.

gap heatmap quantifies the disparity between average and worst-group performance, with MAE showing substantially larger gaps.

C Additional Visualizations

Figure 8 presents a component ranking summary showing the relative impact of different architectural and hyperparameter choices across datasets. Data augmentation consistently emerges as the most impactful component for SimCLR robustness, particularly on Fashion-Pattern where it provides the largest improvement (31.2%).

Figure 9 shows additional ablation studies including normalization, pooling strategy, and augmentation stochasticity effects. Normalization provides substantial benefits on Fashion-Pattern (33.7% improvement) but minimal impact on other datasets. Pooling strategy and augmentation stochasticity show smaller, dataset-dependent effects.

Figure 7: Cross-dataset performance summary heatmaps comparing SimCLR and MAE. Top-left: Worst-group accuracy (higher is better, 0–1 scale). Top-right: SFRS (lower is better, 0–1 scale). Bottom-left: Average accuracy (0–1 scale). Bottom-right: Robustness gap (lower is better, 0–1 scale). SimCLR demonstrates consistently better robustness metrics across CIFAR10-BG and Fashion-Pattern, while both methods completely fail on Color-MNIST.

Component Ablation Summary: Key Findings

Component	Dataset	Effect on Worst-Group	Effect on SFRS	Recommendation
Data Augmentation	CIFAR10-BG	+0.040	-0.05	Essential for robustness
	Fashion-Pattern	+0.312	N/A	Critical for local patterns
Projection Head	CIFAR10-BG	+0.06	-0.06	Beneficial for SimCLR
	Fashion-Pattern	+0.14	-0.14	Strongly recommended
Embedding Dim 256	CIFAR10-BG	Best for SimCLR	Best for SimCLR	Larger is better
Embedding Dim 64	Fashion-Pattern	Optimal	Optimal	Dataset-dependent
Temperature T=1.0	CIFAR10-BG	+0.03	-0.05	Slight improvement
	Fashion-Pattern	-0.11	+0.02	Harmful
MAE Masking 75%	Fashion-Pattern	+0.088	-0.086	Improves robustness
	CIFAR10-BG	+0.013	-0.024	Modest benefit
Remove Pooling	Fashion-Pattern	-0.207 (SimCLR)	+0.195	Harmful for SimCLR
	Color-MNIST	+0.085 (MAE)	-0.15	Helps MAE slightly

Figure 8: Component ranking summary showing the relative impact magnitude of different ablations across datasets. Data augmentation emerges as the most critical component for robustness, particularly on Fashion-Pattern, while other components show dataset-dependent effects.

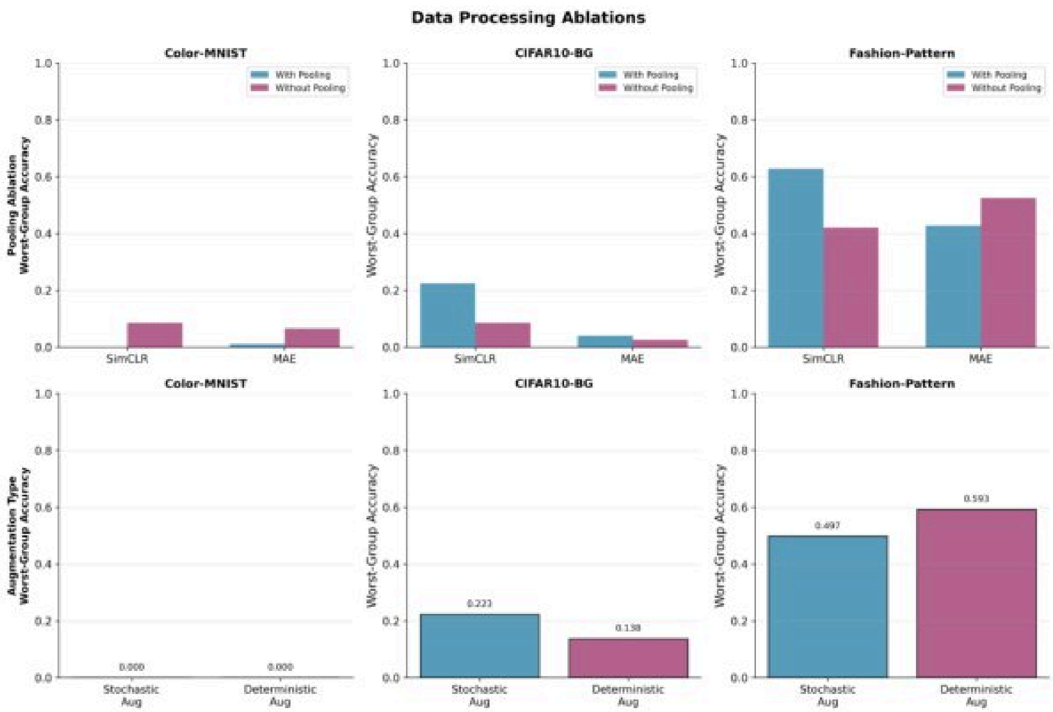

Figure 9: Additional ablations on data processing components. Normalization shows strong dataset-dependent effects with largest benefits on Fashion-Pattern. Pooling strategy and augmentation stochasticity show smaller, more varied effects across datasets.